CURRICULUM VITAE UPDATED 31 OCTOBER 2026

Youngkwon Ko

AI / ML research & software development

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01 Education

Yonsei University March 2023 - Present
B.S. in Artificial Intelligence, Seoul, Republic of Korea.
Currently on a leave of absence while serving in the Republic of Korea Navy.

02 Research interests

Representation learning, computer vision, efficient AI systems, and quantum optimization.

03 Publications

01 **A Mission-Aware Autoencoder Bottleneck Compression Framework for Drone-Based Hyperspectral Image Transmission.**

Youngkwon Ko, Yunbo Shim, Doun Lee, and Seayul Choi.

Marine Science & Technology, 2026.

02 **Provenance-Preserving Schema and Resource Construction for Information Extraction from Historical Battle Records: A Corpus Study of the Inchon Landing.**

Yunbo Shim, **Youngkwon Ko**, et al.

Proceedings of the Korea Computer Congress (KCC 2026), 2026.

RESEARCH MANUSCRIPT

03 **Design and Preliminary Validation of a Public-Data-Based Ship-Like Multi-CCTV Fire and Smoke Early Warning System.**

Yunbo Shim, **Youngkwon Ko**, Doun Lee, Seayul Choi, and Muhyeon Park.

Public-data-based multi-camera detection, temporal verification, and alert evidence capture.

04 Research in progress

PACT Phase-aware compression and transfer for continuing CV-QAOA optimization through short coherent blocks.

Lineage Evidence-grounded resolution of parent relationships between entities.

05 Selected software

P2I / PyTorch model analysis Runtime tracing, model exploration, and an editable ArchitectureIR with an agent-facing harness.

06 Research & professional experience

Republic of Korea Navy December 2025 - Present

Artificial Intelligence Specialist ▪ Expected completion: August 2027

- Developed mission-aware hyperspectral image transmission using autoencoder compression and spectral patch prioritization. Reported approximately 613× and 710.5× latent-payload reductions on KSC and Salinas, respectively.
- Participating in technical research and discussions on hyperspectral satellite systems in collaboration with the Agency for Defense Development (ADD).

Root Education September - December 2025

Co-Founder

- Co-founded an AI education platform organizing admissions-focused influencer content into a structured learning environment.
- Developed AI-assisted learning features and contributed to product design and frontend development with Next.js and TypeScript.

Little Samo September - October 2025

Contract Software Developer

- Built SamoAI-InstagramTracker for AI-powered influencer discovery and data collection on the SamoAI multi-agent framework.
- Contributed to the SamoAI engine and its CLI-based agent chat interface.

Yonsei Medisys Lab January - June 2025

Research Intern

- Researched low-dose CT quality assessment using 11,200 MOS-annotated images; trained and evaluated EfficientNet, ViT, TinyViT, and ManiQA.
- Investigated CT denoising fine-tuning using an image quality assessment model as a reward signal.

Creai+it September 2024 - December 2025

Co-Leader, AI Startup Club

- Led AI product development across document intelligence, legal AI, assistive technology, and generative AI, including Somni and Miney.
- Collaborated with Upstage on a Solar-based document processing system and participated in the 2025 AGI Agent Hackathon.

07 Awards & honors

1st Place, Navy & Marine Corps Artificial Intelligence Competition 2026

RED S Design Award - 1st Prize June 2025

Interactive Media Design, Yonsei University.

08 Technical skills

Programming	Python, TypeScript, C++, JavaScript
Machine learning	PyTorch, TensorFlow, OpenCV
Development	React, Next.js, Git, Linux